



MY TECHNICAL REPORT ON MYWORK EXPERIENCE AT ipNX NIGERIA LIMITED



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Introduction to SIWES

- **What is SIWES?** A program designed to transition theoretical academic knowledge into practical, real-world experience.
- **Role of the ITF:** The scheme was initiated in 1973 by the Industrial Training Fund (ITF), which is responsible for the administration and funding to promote skill development.
- **My Program Duration:** 24 weeks
- **Primary Goal:** To apply classroom theories to real-world scenarios, gain hands-on experience, and enhance employability upon graduation.



Host Company: ipNX Nigeria Limited

- **Who they are:** A leading Nigerian ICT company providing Fibre-to-the-Home (FTTH), enterprise and managed services.
- **My Department:** Retail/Engineering Department.
- **Core Business:** Deploying high-speed internet and voice services.
- **Coverage Areas:** Operated Across Lagos, Abuja, Port-Harcourt, Kano, and Ibadan.
- **Industry Leader:** First company in Nigeria to deploy Fibre-to-the-Home (FTTH) on a significant scale.
- **Recent Recognition:** Won "Most Innovative Digital Connectivity Company" (2024) and "Residential Service Provider of the Year" (Dec 2024)





My Role and Key Responsibilities

- Assisted in FoS (FTTH) installations and Optical Network Terminal (ONT) setup.
- Configured Customer Premises Equipment (CPE), including modems and mesh devices (Wi-Fi SSI).
- Performed network diagnostics and troubleshooting for customer complaints.
- Monitored link performance using the Basic Diagnostic Tool (BDT).
- Handled IP blacklist checks and server accessibility tests.



Internship Metrics & Achievements

100+

TROUBLE TICKETS
RESOLVED

40+

PREVENTIVE SITE VISITS

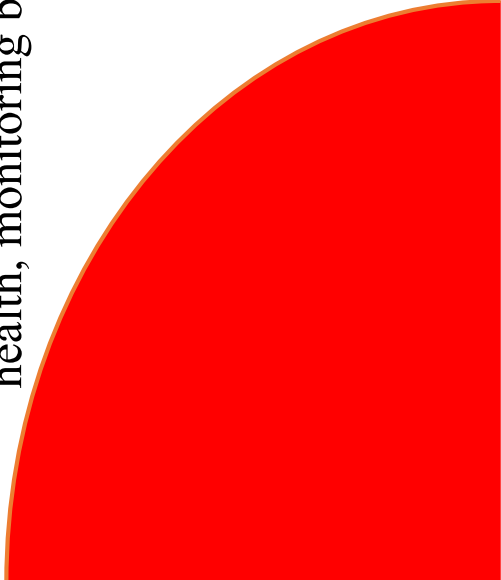
3+

FULL FTTH
DEPLOYMENTS

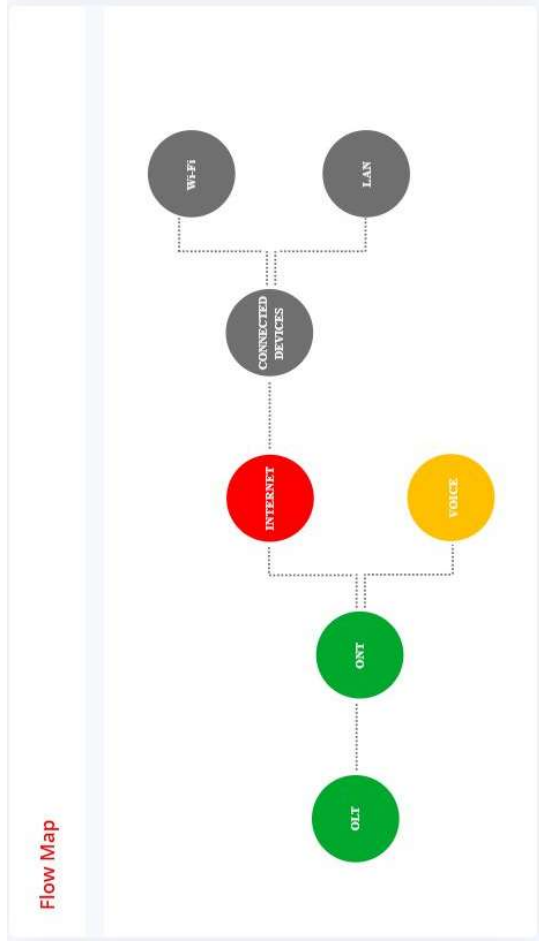
DBN

Technical Experience: Diagnostic Tools

- **Basic Diagnostic Tool (BDT):** Used to view ONT flow maps, check signal quality (RX power), and high losses.
- **Optical Multi-meter (OMM):** To get precise optical power readings (in dBm) and troubleshoot far
- **Wi-Fi Analyzer:** To check for signal interference and find the best Wi-Fi channel.
- **Speedtest.net / Wireshark:** To confirm internet speed and analyse network packets.
- **Mxtoolbox:** To diagnose and troubleshoot network and email infrastructure issues, including check health, monitoring blacklists, and analyzing email delivery problems



Diagnostic Tools Cont'd



Technical Experience: FTTH Installation

Site Survey: Planning and planting the fibre route from the distribution point (MST) to the home.



Fibre Splicing: Terminating the cable to ensure low signal loss.

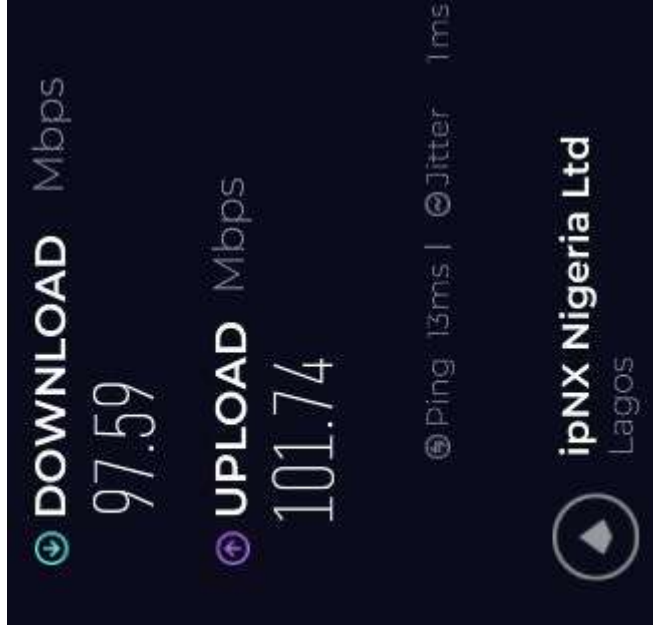


ONT Mounting: Installing and securing the customer's device.



Case Study I: Persistent WAN Disconnect

- **Problem:** Customer had intermittent disconnections and high ping.
- **Diagnosis:** Used OMM and found a catastrophic signal loss. RX Power was **-70.00 dBm** (acceptable range is -28 to -42 dBm).
- **Action:** Escalated to the Operations & Maintenance (O&M) team. A faulty fibre section was re-spliced.
- **Result:** Signal improved to **-21.10 dBm**. Service was restored.



Case Study II: Wi-Fi Optimization

- **Problem:** Customer complained of slow internet, but isolated wired speed tests were fine.
- **Diagnosis:** Used a **Wi-Fi Analyzer** app and found their 2.4Ghz network was on a crowded channel (Channel 6).
- **Action:** Logged into their modem and changed the 2.4Ghz channel to a clear one (Channel 10).
- **Result:** Customer's wireless device speeds improved immediately. This showed the problem wasn't the fiber, it was interference.



Skills Acquired

Technical Skills	Professional (Soft) Skills
Fibre Optic Installation & Splicing	Problem-Solving & Critical Thinking
Modem/Router Configuration (Wi-Fi, DHCP)	Customer Communication & Support
Network Diagnostics (BDT, OMM, Wi-Fi Analyzer)	Teamwork & Collaboration (with O&M teams)
Understanding FTTH, ONTs, and VLANs	Time Management

Technical Concepts Learned

- Understanding of GPON Architecture OLT (Cabinet) → Closure → MST (Splitters) → ONT
- VLAN Tagging and Customer Segmentation
- PPPoE Authentication & Session Management
- BRAS (AAA): Authentication, Authorization, Accounting
- IP Addressing (Public vs Private IPs)
- Latency, Ping, and Traceroute for Network Diagnosis
- Fibre Power Levels (dBm) and Link Stability
- Wi-Fi Channel Interference and Optimization

Challenges Encountered

- Diagnosing intermittent faults that disappeared during testing.
- Handling frustrated customers and managing expectations.
- Dealing with older customer devices that had compatibility issues (e.g., with WPA2 security).
- Environmental challenges during fieldwork (dust, rain).



Conclusion & Recommendations

- **Conclusion:** The internship was a transformational experience that bridged the gap between the real-world application, especially in FTTH and network diagnostics.
 - **Recommendations (for Future Interns):**
 - Be proactive and ask questions.
 - Document your daily activities.
 - Focus on connecting classroom concepts to field tasks.
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Thank You